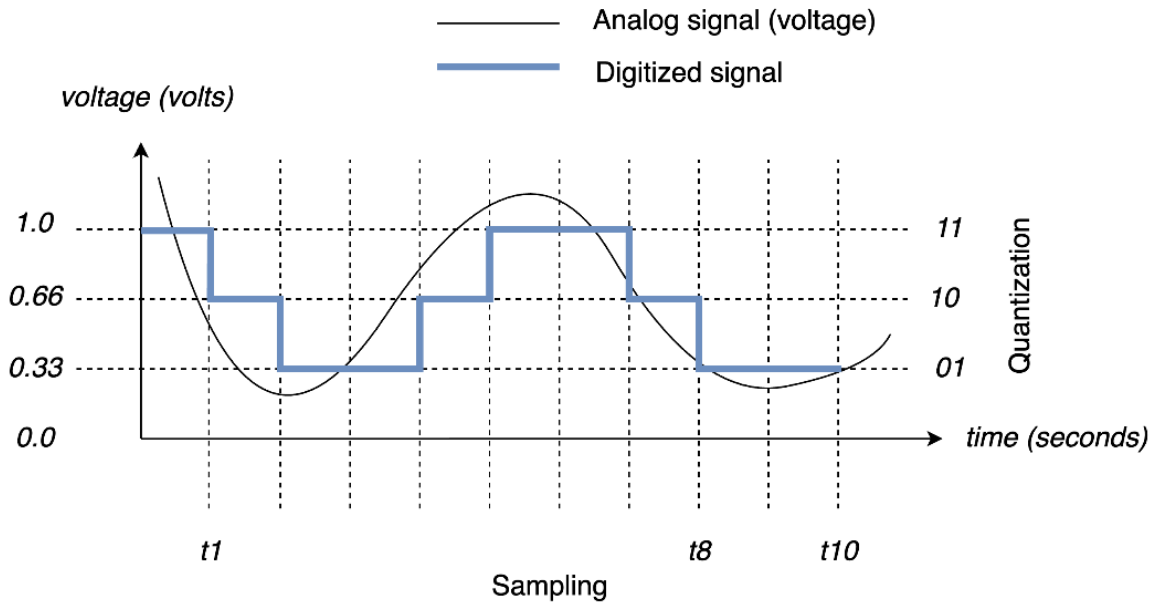
HOW IS A PHYSICAL CHANGE CONVERTED TO A DIGITAL VALUE?

The analog to digital conversion at the core of the sensor signal chain is illustrated below. The continuously varying voltage caused by temperature fluctuations is *sampled* once every second (or at a frequency of 1Hz), and is *quantized* with a 2-bit converter. *Quantization* refers to the process of mapping the continuous output value (the voltage from the sensor), to a smaller set of values. In a 2-bit converter, the continuous voltage is mapped to 2^2 (4) possible values represented by 2 bits. The figure shows that mapping the continuous voltage to one of 4 possible values every second means that some information is lost between time ***t8*** and time ***t10***, where the fall in voltage below 0.33V is not translated to the digital signal. If it's important to your digital twin to be able to detect that fall in temperature, you need to use a sensor with different conversion characteristics.



Listing 3.1 shows how analog to digital conversion works to convert a voltage reading to a calibrated moisture percentage in a 10 bit converter.